

# MomCare LK

## Project Proposal

*Sri Lanka's first AI-powered all-in-one web platform for busy working pregnant mothers*

|               |                                                        |
|---------------|--------------------------------------------------------|
| Platform      | Web Application (PWA)                                  |
| Target Market | Working pregnant mothers in Sri Lanka                  |
| Team Size     | 5 Members                                              |
| Deadline      | Sunday, June 28, 2026                                  |
| Submission    | Add group details & topic to LMS Google Sheet (Week 3) |

## 1. Project Overview

MomCare LK is Sri Lanka's first AI-powered all-in-one web platform designed specifically for busy working pregnant mothers. It combines smart clinic reminders, one-touch emergency response, nearby baby shop discovery, and emotional wellbeing support — all available in Sinhala, Tamil, and English.

### Why this idea?

- Over 350,000 pregnancies per year in Sri Lanka
- A growing number of mothers work full-time during pregnancy
- They juggle office deadlines, clinic appointments, emergency readiness, and baby shopping simultaneously
- Zero dedicated digital platforms exist in Sri Lanka for this specific need
- Real emotional and practical value — helps mothers and families island-wide

## 2. Core Features — Four Pillars

### Pillar 1 — Health & Clinic Management

- Smart clinic appointment reminders (SMS + browser notifications)
- Vaccination, blood test, and scan tracking by pregnancy week
- Week-by-week pregnancy tracker with baby development updates
- AI health Q&A; assistant — available 24/7 in Sinhala, Tamil, and English
- Digital health record storage (blood pressure, weight, scan reports, doctor notes)
- One-click report sharing with doctors or family members

### Pillar 2 — Emergency & Safety

- One-touch SOS button connected to Sri Lanka's 1990 Suwa Seriya ambulance
- Automatic location detection and sharing of pregnancy week and health details
- Simultaneous emergency alert to all pre-saved contacts
- Nearest maternity hospital finder with 24-hour availability filter
- Directions and contact numbers for hospitals
- Safe travel mode — periodic check-ins; auto-alerts contacts if mom stops responding

### Pillar 3 — Shopping & Baby Products

- GPS-based nearby baby shop, pharmacy, and supermarket finder
- Filter by product type (diapers, formula, clothing, medicine)
- Smart pregnancy shopping checklist organised by trimester
- Price comparison for common baby products across local Sri Lankan online stores
- Community buy-and-sell board for gently used baby items
- Supports low-income families and reduces waste

#### **Pillar 4 — Emotional Wellbeing & Community**

- Daily mood and wellbeing check-in
- AI detection of prenatal depression and anxiety patterns
- Working mom stress manager — tracks work/rest balance
- Gentle break reminders (e.g. 'You haven't rested in 3 hours')
- MomConnect community forum with topic-based discussions
- Verified midwife and doctor accounts answer questions in forum
- Partner and family dashboard — clinic dates, pregnancy week, health updates, SOS alerts

### 3. Full Feature Overview

| Feature Area       | Details                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------|
| Health & Clinic    | Clinic reminders, vaccination tracking, pregnancy week tracker, AI health Q&A, digital records |
| Emergency & Safety | SOS button (1990), location share, hospital finder, safe travel mode, family alerts            |
| Shopping           | Baby shop finder, trimester checklist, price comparison, community buy & sell                  |
| Wellbeing          | Mood tracker, burnout detection, stress manager, community forum, partner dashboard            |
| AI Assistant       | GPT-4o powered, trilingual (Sinhala/Tamil/English), 24/7 availability                          |
| Safe Travel Mode   | Periodic check-ins, auto-alert if silent, partner notification                                 |
| Community Forum    | Topic-based discussions, verified midwife/doctor accounts                                      |
| Partner Dashboard  | Clinic dates, pregnancy week, health updates, SOS alert visibility                             |
| Payment            | PayHere integration — cards, bank transfer, eZCash/mCash                                       |
| Accessibility      | Progressive Web App (PWA) — works like a mobile app on any phone browser                       |

### 4. User Account Types

| Account Type              | Access & Purpose                                                         |
|---------------------------|--------------------------------------------------------------------------|
| Mom Account               | Primary user — full platform access across all four pillars              |
| Partner / Husband Account | View clinic dates, pregnancy week, health updates, and SOS alerts        |
| Doctor / Midwife Account  | Monitor patients, respond in community forum, view shared health records |

### 5. Innovation Layer

| Innovation                 | Description                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| AI Health Assistant        | GPT-4o powered chatbot answering health questions in Sinhala, Tamil, and English — 24/7                           |
| Safe Travel Mode           | Mom activates it when traveling alone; auto-alerts contacts if she stops responding                               |
| Burnout & Stress Detection | Tracks study hours, rest gaps, mood check-ins; flags prenatal depression patterns to suggest professional support |
| PayHere Payment Gateway    | Sri Lanka's most popular gateway; supports cards, bank transfer, eZCash, mCash                                    |
| Trilingual Support         | Full Sinhala, Tamil, and English UI and AI responses — first in Sri Lanka for pregnancy apps                      |

| Innovation           | Description                                                                       |
|----------------------|-----------------------------------------------------------------------------------|
| PWA Mobile Support   | Web app works like a native mobile app on any phone browser — no app store needed |
| Community Buy & Sell | Mothers sell gently used baby items to other mothers, reducing cost and waste     |
| Partner Dashboard    | Husbands and family members stay informed and connected to the pregnancy journey  |

## 6. Technical Stack

### Frontend

| Tool / Technology | Purpose                                                       |
|-------------------|---------------------------------------------------------------|
| React.js          | Main frontend framework                                       |
| Tailwind CSS      | Fast, clean, responsive UI styling                            |
| Figma             | UI/UX wireframe and prototype design                          |
| Axios             | API calls between frontend and backend                        |
| PWA               | Progressive Web App — mobile-app experience without app store |

### Backend

| Tool / Technology | Purpose                                           |
|-------------------|---------------------------------------------------|
| Node.js + Express | Server and REST API                               |
| JWT               | Secure user authentication and session management |
| Socket.io         | Real-time emergency alerts and notifications      |
| REST API          | Communication between all platform modules        |

### Database & Storage

| Tool / Technology | Purpose                                           |
|-------------------|---------------------------------------------------|
| MySQL / Supabase  | Main relational database for user and health data |
| Cloudinary        | Scan report and document file storage             |
| Firebase          | Real-time push notifications                      |

### AI & Integrations

| Tool / Technology      | Purpose                                            |
|------------------------|----------------------------------------------------|
| OpenAI GPT-4o          | Core AI health Q&A; assistant and stress detection |
| Google Maps API        | Hospital, pharmacy, and baby shop finder           |
| Twilio / Dialog API    | SMS reminders and emergency alerts                 |
| Google Translate API   | Sinhala, Tamil, and English language support       |
| Web Push Notifications | Clinic date and appointment reminders              |

Hosting & Deployment

| Tool / Technology | Purpose                                                         |
|-------------------|-----------------------------------------------------------------|
| Vercel            | Frontend hosting (free tier available)                          |
| Railway / Render  | Backend hosting (free tier available)                           |
| Supabase          | Database hosting (free tier available)                          |
| GitHub            | Version control and team collaboration                          |
| PayHere           | Sri Lanka payment gateway — cards, bank transfer, eZCash, mCash |

## 7. Cost Estimation

### Monthly Running Costs

|                              |                                         |
|------------------------------|-----------------------------------------|
| Hosting (Vercel + Railway)   | Free – LKR 5,000                        |
| Database (Supabase)          | Free – LKR 3,000                        |
| Google Maps API              | LKR 2,000 – 8,000                       |
| OpenAI API (GPT-4o)          | LKR 5,000 – 15,000                      |
| SMS Alerts (Twilio / Dialog) | LKR 3,000 – 10,000                      |
| Domain (.lk)                 | ~LKR 400 (monthly split of annual cost) |
| Storage (Cloudinary)         | Free – LKR 2,000                        |
| TOTAL ESTIMATED              | LKR 10,400 – 43,400 / month             |

Note: As a student project, costs can be significantly reduced by using free tiers of Vercel, Supabase, Cloudinary, and Railway during development and early launch.

## 8. Pricing Plans

| Package              | Price                       | Key Features                                                                 |
|----------------------|-----------------------------|------------------------------------------------------------------------------|
| Free                 | LKR 0                       | Clinic reminders, week tracker, emergency SOS button                         |
| Mom Basic            | LKR 490 / month             | Full past paper library, AI health Q&A, baby shop finder, shopping checklist |
| Mom Premium          | LKR 990 / month             | Mood tracker, burnout detection, partner dashboard, community forum          |
| Family Plan          | LKR 1,290 / month           | Full family access, safe travel mode, digital health records                 |
| School / Clinic Plan | LKR 15,000 – 25,000 / month | Up to 50 accounts, doctor/midwife panel, class-wide analytics                |

### Revenue Potential Estimate

|                           |                      |
|---------------------------|----------------------|
| 100 moms on Mom Basic     | ~LKR 49,000 / month  |
| 500 moms on mixed plans   | ~LKR 400,000 / month |
| 5 clinic / hospital plans | ~LKR 100,000 / month |
| Combined realistic target | ~LKR 500,000 / month |



## 9. Target Audience

| Audience                  | Why They Need MomCare LK                                                |
|---------------------------|-------------------------------------------------------------------------|
| Working pregnant mothers  | Primary — busy, need reminders, emergency support, and health guidance  |
| First-time mothers        | Anxious and need continuous guidance throughout pregnancy               |
| Husbands & family members | Want to stay informed and provide support safely                        |
| Midwives & doctors        | Monitor patients and advise through the community platform              |
| Rural mothers             | Need hospital finder and health guidance most — limited local resources |
| Low-income families       | Benefit from price comparison and community buy & sell features         |

### Geographic Focus

- Start with Western Province (Colombo, Gampaha, Kalutara) — highest working-mother density
- Expand island-wide, prioritising semi-urban and rural areas with limited healthcare access
- Full Sinhala and Tamil support ensures reach beyond English-speaking urban population

## 10. Marketing Strategy

- Social media marketing on Facebook, TikTok, and YouTube — most-used platforms by Sri Lankan mothers and families
- Partner with maternity clinics, hospitals, and midwives to recommend the platform to patients
- Free tier strategy — hook users with free clinic reminders and emergency SOS, then convert to paid
- Sinhala and Tamil language marketing — massively increases trust and reach among non-English speakers
- Referral program — 'Refer a friend and get 1 month free' to leverage word of mouth
- WhatsApp broadcast lists targeting mother communities and parenting groups
- Collaborate with Sri Lankan parenting bloggers and health influencers on YouTube and Instagram
- Offer yearly subscription discounts (e.g. Mom Premium yearly at LKR 8,900 vs LKR 11,880 monthly)

## 11. Topic Approval Criteria

|                      |                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| ✓ Innovative         | AI health assistant with emotion/stress detection, safe travel mode, and community buy & sell — first of their kind in Sri Lanka |
| ✓ Gap-Filling        | No dedicated pregnancy companion platform exists in Sri Lanka; tuition classes are expensive and inaccessible for many           |
| ✓ Sri Lankan Context | Integrates 1990 Suwa Seriya ambulance, local baby shops, PayHere payment, and Sinhala/Tamil/English trilingual support           |
| ✓ Timely             | Growing working-mother population, post-economic crisis family budget pressure, and rising smartphone/internet penetration       |

✓ Platform

Web Application with PWA support — accessible on any phone browser without an app store download

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